5CCSAMLF Coursework 2 Report

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1 Introduction

The paper studies active learning in the low-budget setting, where only a very small number of samples can be labelled. It argues that uncertainty-based methods perform poorly in this regime and proposes TypiClust/TPCRP, which instead selects representative and typical samples from the data distribution. The aim is to improve performance when labelled data is extremely limited.

2 Methodology

TPCRP works by mapping images into a feature space, clustering them to maintain diversity, and selecting the most typical point from each chosen cluster using local neighbourhood distance. In this coursework, the method was evaluated on CIFAR-10 using Framework 1 (fully supervised training on queried samples), Framework 2 (ResNet-18 embeddings with a linear classifier), and a simplified Framework 3 semi-supervised approximation. Performance was compared against Random, Margin, and Entropy baselines.

3 Results

Framework	Method	10	20	30	40	50	60
F1	TPCRP	0.1053	0.1146	0.1323	0.1064	0.1003	0.1503
	Random	0.0971	0.1001	0.1044	0.1513	0.1011	0.1000
	Margin	0.1189	0.1254	0.1032	0.1196	0.1220	0.1031
	Entropy	0.0984	0.0987	0.1189	0.1226	0.1402	0.1146
F2	TPCRP	0.5349	0.6411	0.6556	0.6699	0.6695	0.6793
	Random	0.3848	0.5309	0.6066	0.6302	0.6441	0.6468
	Margin	0.3774	0.5813	0.6344	0.6567	0.6674	0.6652
	Entropy	0.3704	0.3861	0.4934	0.5315	0.5334	0.5641
F3	TPCRP	0.1187	0.1037	0.1329	0.1149	0.0933	0.1001
	Random	0.0857	0.1072	0.1460	0.0863	0.0975	0.0998
	Margin	0.1005	0.1045	0.1021	0.1045	0.1000	0.1011
	Entropy	0.0994	0.1094	0.1000	0.0789	0.1000	0.1001

Table 1: Accuracy across label budgets.

4 Plots

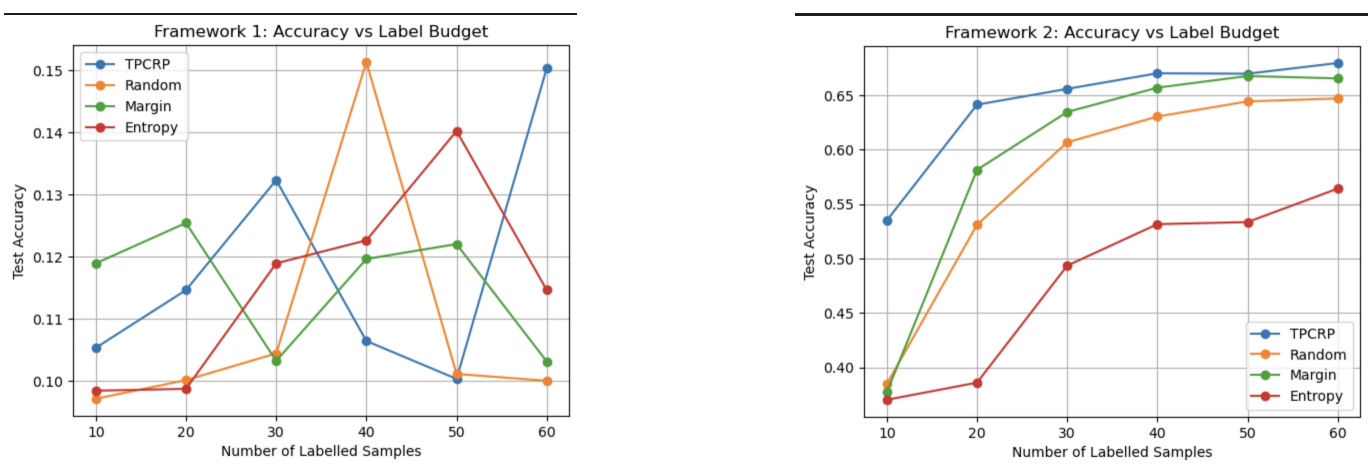
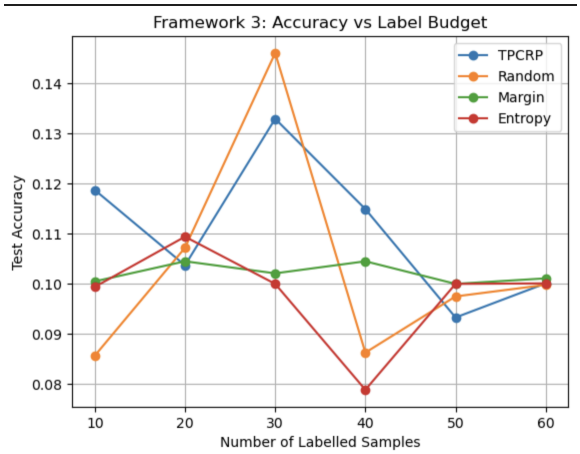


Figure 1: Left: Framework 1 performance across methods. Right: Framework 2 performance across methods



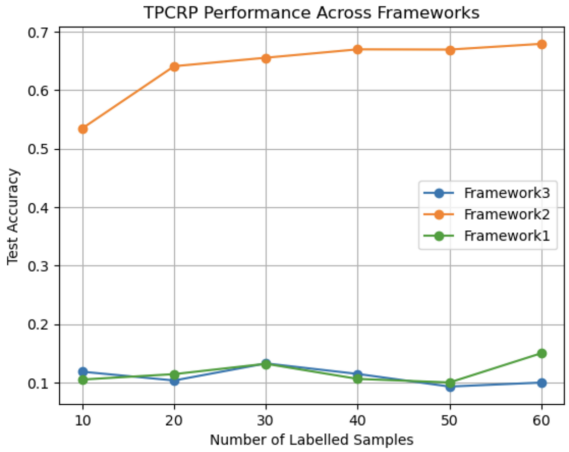


Figure 2: Left: Framework 3 performance across methods. Right: TPCRP across frameworks.

5 Statistical Analysis

Experiments were run once per method and budget, so the analysis focuses on comparative trends rather than formal significance testing. Framework 2 produced the strongest results overall. TPCRP was the best-performing method at every tested budget, increasing from 53.49% at 10 labels to 67.93% at 60 labels, and outperforming Random, Margin, and Entropy throughout. The largest gains appeared at the smallest budgets, which is consistent with the paper's claim that representative and typical samples are especially useful in the low-budget regime.

By contrast, Framework 1 remained close to chance level for all methods, suggesting that training a deep network from scratch on extremely small labelled subsets was unstable. Framework 3 was also weak overall, with all methods staying close to 10%–15%, likely because the semi-supervised stage was implemented as a simplified approximation rather than a full FlexMatch reproduction. Across all three frameworks, the strongest and clearest support for TPCRP was therefore observed in Framework 2.

6 Modification

I modified TPCRP by incorporating an entropy-based uncertainty term into the within-cluster scoring rule. The aim was to preserve the representativeness and diversity of the original TPCRP method while also favouring informative samples for the classifier.

The modified method was evaluated in Framework 2. It did not outperform TPCRP at every budget, but it improved performance at several later rounds and achieved the best final result overall. In particular, accuracy improved from 65.56% to 66.00% at 30 labels, from 66.95% to 68.39% at 50 labels, and from 67.93% to 69.35% at 60 labels. This suggests that the modification was most beneficial once a slightly larger labelled set had been acquired.

7 Conclusion

In conclusion, the results show that TPCRP was most effective in Framework 2, where it consistently outperformed Random, Margin and Entropy across all tested budgets. This supports the paper's main claim that in the low-budget regime, representative and diverse sample selection is more effective than standard uncertainty-based querying. Framework 1 and Framework 3 remained much weaker overall, showing that the choice of training framework had a major impact on final performance.

A key limitation of this coursework was computational cost. Because of this, only the most practical baselines to implement and compare were included in the final experiments: **Random**, **Margin**, and **Entropy**. Similarly, the original SimCLR-based representation learning stage used by the paper for TPCRP on CIFAR-10. was not fully retained in the final implementation, and the semi-supervised framework was approximated rather than fully reproduced with FlexMatch. Nevertheless, the overall results still provide evidence that TPCRP is a strong strategy in the low-budget setting, especially when combined with a learned feature embedding.

8 Code Supplement

8.1 Code Printout

In [63]:
```python
import torch
from torchvision import datasets, transforms
import torch.optim as optim
import matplotlib.pyplot as plt
import torch.nn as nn
import torch.nn.functional as F
from torchvision.models import resnet18
from torch.utils.data import DataLoader, TensorDataset, Dataset, Subset
import pandas as pd
import numpy as np
from sklearn.cluster import KMeans
from sklearn.metrics import pairwise_distances
import random
from torchvision.datasets import CIFAR10
```

In [2]:
```python
trainTransform = transforms.Compose([transforms.ToTensor()])
testTransform = transforms.Compose([transforms.ToTensor()])

trainSet = CIFAR10(root="./data", train=True, download=True, transform=trainTransform)
testSet = CIFAR10(root="./data", train=False, download=True, transform=testTransform)
trainLoader = DataLoader(trainSet, batch_size=128, shuffle=True, num_workers=0)
testLoader = DataLoader(testSet, batch_size=128, shuffle=False, num_workers=0)
```

```
Files already downloaded and verified
Files already downloaded and verified
```

In [3]:
```python
class ResNet18(nn.Module):
    def __init__(self, num_classes=10):
        super(ResNet18, self).__init__()
        backbone = resnet18(weights=None)
        backbone.conv1 = nn.Conv2d(in_channels=3, out_channels=64, kernel_size=3, stride=1, padding=1, bias=Fals
        backbone.maxpool = nn.Identity()
        self.feature_extractor = nn.Sequential(*list(backbone.children())[:-1])
        self.classifier = nn.Linear(512, num_classes)

    def forward(self, x):
        features = self.feature_extractor(x)
        features = torch.flatten(features, 1)
        output = self.classifier(features)
        return output

    def getEmbeddings(self, x):
        features = self.feature_extractor(x)
        features = torch.flatten(features, 1)
        features = F.normalize(features, p=2, dim=1)
        return features
```

In [4]:
```python
class ResNet18Classifier(nn.Module):
    def __init__(self, numClasses=10):
        super().__init__()
        self.backbone = resnet18(weights=None)
        self.backbone.fc = nn.Linear(self.backbone.fc.in_features, numClasses)

    def forward(self, x):
        return self.backbone(x)
```

In [23]:
```python
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(device)

def trainResNetClassifier(model, trainLoader, numEpochs=10):
    model = model.to(device)
    lossFn = nn.CrossEntropyLoss()
    optimizer = optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-4)

    for epoch in range(numEpochs):
        model.train()
        runningLoss = 0.0
        correct = 0
        total = 0

        for images, labels in trainLoader:
            images = images.to(device)
            labels = labels.to(device)

            optimizer.zero_grad()
            logits = model(images)
            loss = lossFn(logits, labels)
            loss.backward()
            optimizer.step()
```

```
            runningLoss += loss.item()
            preds = torch.argmax(logits, dim=1)
            correct += (preds == labels).sum().item()
            total += labels.size(0)

        epochLoss = runningLoss / len(trainLoader)
        epochAcc = correct / total
        print(f"Epoch {epoch + 1}/{numEpochs}, Loss: {epochLoss:.4f}, Train Acc: {epochAcc:.4f}")

    return model
```

```
cpu
```

In [24]:
```
def evaluateClassifier(model, testLoader):
    model.eval()
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in testLoader:
            images = images.to(device)
            labels = labels.to(device)

            logits = model(images)
            preds = torch.argmax(logits, dim=1)

            correct += (preds == labels).sum().item()
            total += labels.size(0)

    testAccuracy = correct / total
    return testAccuracy
```

In [25]:
```
resnetModel = ResNet18Classifier(numClasses=10).to(device)
resnetModel = trainResNetClassifier(resnetModel, trainLoader, numEpochs=10)

testAcc = evaluateClassifier(resnetModel, testLoader)
print("Full ResNet test accuracy:", testAcc)

torch.save(resnetModel.state_dict(), "resnet18Cifar10.pth")
```

```
Epoch 1/10, Loss: 1.3685, Train Acc: 0.5062
Epoch 2/10, Loss: 0.9853, Train Acc: 0.6510
Epoch 3/10, Loss: 0.8088, Train Acc: 0.7139
Epoch 4/10, Loss: 0.6914, Train Acc: 0.7559
Epoch 5/10, Loss: 0.5959, Train Acc: 0.7915
Epoch 6/10, Loss: 0.5100, Train Acc: 0.8209
Epoch 7/10, Loss: 0.4386, Train Acc: 0.8459
Epoch 8/10, Loss: 0.3746, Train Acc: 0.8691
Epoch 9/10, Loss: 0.3231, Train Acc: 0.8877
Epoch 10/10, Loss: 0.2654, Train Acc: 0.9079
Full ResNet test accuracy: 0.6518
```

In [38]:
```
class ResNet18FeatureExtractor(nn.Module):
    def __init__(self, trainedModel):
        super().__init__()
        self.features = nn.Sequential(*list(trainedModel.backbone.children())[:-1])

    def forward(self, x):
        x = self.features(x)
        x = torch.flatten(x, 1)
        return x
```

In [39]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
resnetModel = ResNet18Classifier(numClasses=10).to(device)
resnetModel.load_state_dict(torch.load("resnet18Cifar10.pth", map_location=device))
resnetModel.eval()
featureExtractor = ResNet18FeatureExtractor(resnetModel).to(device)
featureExtractor.eval()
```

```
/var/folders/h4/l90wkg5j0dz27gsv50bz6f000000gn/T/ipykernel_74979/2245030857.py:3: FutureWarning: You are using `
torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicit
ly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See h
ttps://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release,
the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed
during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explic
itly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights
_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHu
b for any issues related to this experimental feature.
  resnetModel.load_state_dict(torch.load("resnet18Cifar10.pth", map_location=device))
```

Out[39]:
```
ResNet18FeatureExtractor(
  (features): Sequential(
    (0): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3, 3), bias=False)
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
```

```
    (2): ReLU(inplace=True)
    (3): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1, ceil_mode=False)
    (4): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
      (1): BasicBlock(
        (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (5): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(64, 128, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(2, 2), bias=False)
          (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (6): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(2, 2), bias=False)
          (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (7): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), bias=False)
          (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (8): AdaptiveAvgPool2d(output_size=(1, 1))
  )
)
```

In [40]:

```
def extract_embeddings(model, dataset):
```

```
    loader = DataLoader(dataset, batch_size=256, shuffle=False, num_workers=0)
    model.eval()

    allEmbeddings = []
    allLabels = []

    with torch.no_grad():
        for images, labels in loader:
            images = images.to(device)
            features = model(images)
            allEmbeddings.append(features.cpu())
            allLabels.append(labels)

    allEmbeddings = torch.cat(allEmbeddings, dim=0)
    allLabels = torch.cat(allLabels, dim=0)

    return allEmbeddings, allLabels

trainEmbeddings, trainLabels = extract_embeddings(featureExtractor, trainSet)
testEmbeddings, testLabels = extract_embeddings(featureExtractor, testSet)
torch.save(trainEmbeddings, "train_embeddings.pt")
torch.save(trainLabels, "train_labels.pt")
torch.save(testEmbeddings, "test_embeddings.pt")
torch.save(testLabels, "test_labels.pt")
```

In [41]:

```
def computeTypicality(clusterEmbeddings, kNeighbors=20):
    clusterEmbeddingsNp = clusterEmbeddings.cpu().numpy()
    n = len(clusterEmbeddingsNp)

    if n == 1:
        return np.array([float("inf")])

    kNeighbors = min(kNeighbors, n - 1)

    distMatrix = pairwise_distances(clusterEmbeddingsNp, metric="euclidean")
    np.fill_diagonal(distMatrix, np.inf)

    sortedDists = np.sort(distMatrix, axis=1)
    meanKnnDist = sortedDists[:, :kNeighbors].mean(axis=1)

    typicalityScores = 1.0 / (meanKnnDist + 1e-8)
    return typicalityScores
```

In [29]:

```
def selectInitialTpcrp(trainEmbeddings, unlabeledIndices, budget):
    unlabeledIndices = [int(idx) for idx in unlabeledIndices]
    unlabeledEmbeddings = trainEmbeddings[unlabeledIndices].cpu().numpy()

    kmeans = KMeans(n_clusters=budget, random_state=42, n_init=10)
    clusterLabels = kmeans.fit_predict(unlabeledEmbeddings)

    selectedIndices = []

    for clusterId in range(budget):
        clusterPositions = np.where(clusterLabels == clusterId)[0]

        if len(clusterPositions) == 0:
            continue

        clusterGlobalIndices = [unlabeledIndices[pos] for pos in clusterPositions]
        clusterEmbeddings = trainEmbeddings[clusterGlobalIndices]

        typicalityScores = computeTypicality(clusterEmbeddings)
        bestLocal = int(np.argmax(typicalityScores))
        bestGlobal = clusterGlobalIndices[bestLocal]

        selectedIndices.append(bestGlobal)

    return selectedIndices
```

In [30]:

```
def selectTpcrpBatch(trainEmbeddings, labeledIndices, unlabeledIndices, batchSize):
    labeledIndices = [int(idx) for idx in labeledIndices]
    unlabeledIndices = [int(idx) for idx in unlabeledIndices]

    allIndices = labeledIndices + unlabeledIndices
    allEmbeddings = trainEmbeddings[allIndices].cpu().numpy()

    numClusters = len(labeledIndices) + batchSize
    kmeans = KMeans(n_clusters=numClusters, random_state=42, n_init=10)
    clusterLabels = kmeans.fit_predict(allEmbeddings)

    clusterToMembers = {}
    for pos, clusterId in enumerate(clusterLabels):
        globalIdx = allIndices[pos]
```

```
        if clusterId not in clusterToMembers:
            clusterToMembers[clusterId] = []
        clusterToMembers[clusterId].append(globalIdx)

    uncoveredClusters = []
    for clusterId, members in clusterToMembers.items():
        hasLabeled = any(idx in labeledIndices for idx in members)
        if not hasLabeled:
            uncoveredClusters.append((clusterId, members))

    uncoveredClusters.sort(key=lambda x: len(x[1]), reverse=True)
    chosenClusters = uncoveredClusters[:batchSize]

    selectedIndices = []

    for clusterId, members in chosenClusters:
        unlabeledMembers = [idx for idx in members if idx in unlabeledIndices]

        if len(unlabeledMembers) == 0:
            continue

        clusterEmbeddings = trainEmbeddings[unlabeledMembers]
        typicalityScores = computeTypicality(clusterEmbeddings)
        bestLocal = int(np.argmax(typicalityScores))
        bestGlobal = unlabeledMembers[bestLocal]

        selectedIndices.append(bestGlobal)

    return selectedIndices
```

In [31]:
```
def selectRandomBatch(unlabeledIndices, batchSize):
    return random.sample(unlabeledIndices, batchSize)
```

In [56]:
```
def trainLinearClassifier(trainEmbeddings, trainLabels, selectedIndices):
    selectedIndices = [int(idx) for idx in selectedIndices]

    xTrainSmall = trainEmbeddings[selectedIndices]
    yTrainSmall = trainLabels[selectedIndices]

    trainDatasetSmall = TensorDataset(xTrainSmall, yTrainSmall)
    trainLoaderSmall = DataLoader(
        trainDatasetSmall,
        batch_size=len(trainDatasetSmall),
        shuffle=True
    )

    linearClassifier = nn.Linear(512, 10)
    lossFn = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(linearClassifier.parameters(), lr=0.001)

    numEpochs = 200

    for epoch in range(numEpochs):
        linearClassifier.train()
        for xBatch, yBatch in trainLoaderSmall:
            optimizer.zero_grad()
            logits = linearClassifier(xBatch)
            loss = lossFn(logits, yBatch)
            loss.backward()
            optimizer.step()

    return linearClassifier
```

In [54]:
```
def evaluateLinearClassifier(linearClassifier, testEmbeddings, testLabels):
    linearClassifier.eval()
    with torch.no_grad():
        testLogits = linearClassifier(testEmbeddings)
        testPreds = torch.argmax(testLogits, dim=1)
        testAccuracy = (testPreds == testLabels).float().mean().item()
    return testAccuracy
```

In [55]:
```
def getFramework2Probabilities(linearClassifier, trainEmbeddings, unlabeledIndices):
    unlabeledEmbeddings = trainEmbeddings[unlabeledIndices]

    linearClassifier.eval()
    with torch.no_grad():
        logits = linearClassifier(unlabeledEmbeddings)
        probabilities = torch.softmax(logits, dim=1)

    return probabilities
```

In [ ]:
```
def runFramework2TpcrpLoop(trainEmbeddings, trainLabels, testEmbeddings, testLabels, initialBudget=10, batchSiz
```

```
    labeledIndices = []
    unlabeledIndices = list(range(len(trainEmbeddings)))

    results = []

    initialSelected = selectInitialTpcrp(trainEmbeddings, unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    currentLinear = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)

    accuracy = evaluateLinearClassifier(
        currentLinear,
        testEmbeddings,
        testLabels
    )

    results.append({
        "round": 0,
        "numLabeled": len(labeledIndices),
        "accuracy": accuracy
    })

    for roundNum in range(1, numRounds + 1):
        newSelected = selectTpcrpBatch(
            trainEmbeddings,
            labeledIndices,
            unlabeledIndices,
            batchSize
        )

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]
        newLinear = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)

        accuracy = evaluateLinearClassifier(
            newLinear,
            testEmbeddings,
            testLabels
        )

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

    return results
```

In [ ]:

```
def runFramework2RandomLoop(trainEmbeddings, trainLabels, testEmbeddings, testLabels, initialBudget=10, batchSi
    labeledIndices = []
    unlabeledIndices = list(range(len(trainEmbeddings)))

    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]
    currentLinear = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)

    accuracy = evaluateLinearClassifier(
        currentLinear,
        testEmbeddings,
        testLabels
    )

    results.append({
        "round": 0,
        "numLabeled": len(labeledIndices),
        "accuracy": accuracy
    })

    for roundNum in range(1, numRounds + 1):
        newSelected = selectRandomBatch(unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

        newLinear = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)
        accuracy = evaluateLinearClassifier(
            newLinear,
```

```
            testEmbeddings,
            testLabels
        )

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

    return results
```

In [35]:
```
trainEmbeddings = torch.load("train_embeddings.pt")
trainLabels = torch.load("train_labels.pt")
testEmbeddings = torch.load("test_embeddings.pt")
testLabels = torch.load("test_labels.pt")
```

```
/var/folders/h4/l90wkg5j0dz27gsv50bz6f000000gn/T/ipykernel_74979/1962055859.py:1: FutureWarning: You are using `
torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicit
ly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See h
ttps://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release,
the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed
during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explic
itly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights
_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHu
b for any issues related to this experimental feature.
  trainEmbeddings = torch.load("train_embeddings.pt")
/var/folders/h4/l90wkg5j0dz27gsv50bz6f000000gn/T/ipykernel_74979/1962055859.py:2: FutureWarning: You are using `
torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicit
ly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See h
ttps://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release,
the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed
during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explic
itly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights
_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHu
b for any issues related to this experimental feature.
  trainLabels = torch.load("train_labels.pt")
/var/folders/h4/l90wkg5j0dz27gsv50bz6f000000gn/T/ipykernel_74979/1962055859.py:3: FutureWarning: You are using `
torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicit
ly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See h
ttps://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release,
the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed
during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explic
itly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights
_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHu
b for any issues related to this experimental feature.
  testEmbeddings = torch.load("test_embeddings.pt")
/var/folders/h4/l90wkg5j0dz27gsv50bz6f000000gn/T/ipykernel_74979/1962055859.py:4: FutureWarning: You are using `
torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicit
ly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See h
ttps://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release,
the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed
during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explic
itly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights
_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHu
b for any issues related to this experimental feature.
  testLabels = torch.load("test_labels.pt")
```

In [43]:
```
def trainSupervisedModel(trainSet, labeledIndices, numEpochs=5):
    labeledSubset = Subset(trainSet, labeledIndices)
    labeledLoader = DataLoader(
        labeledSubset,
        batch_size=64,
        shuffle=True,
        num_workers=0
    )

    model = ResNet18Classifier(numClasses=10).to(device)
    lossFn = nn.CrossEntropyLoss()
    optimizer = optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-4)

    for epoch in range(numEpochs):
        model.train()
        runningLoss = 0.0
        correct = 0
        total = 0

        for images, labels in labeledLoader:
            images = images.to(device)
            labels = labels.to(device)

            optimizer.zero_grad()
            logits = model(images)
```

```
            loss = lossFn(logits, labels)
            loss.backward()
            optimizer.step()

            runningLoss += loss.item()
            preds = torch.argmax(logits, dim=1)
            correct += (preds == labels).sum().item()
            total += labels.size(0)

        epochLoss = runningLoss / len(labeledLoader)
        epochAcc = correct / total
        print(f"Epoch {epoch + 1}/{numEpochs}, Loss: {epochLoss:.4f}, Train Acc: {epochAcc:.4f}")

    return model
```

In [44]:

```
def evaluateClassifier(model, testLoader):
    model.eval()
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in testLoader:
            images = images.to(device)
            labels = labels.to(device)

            logits = model(images)
            preds = torch.argmax(logits, dim=1)

            correct += (preds == labels).sum().item()
            total += labels.size(0)

    testAccuracy = correct / total
    return testAccuracy
```

In [46]:

```
def extractResNetEmbeddings(featureExtractor, dataset):
    loader = DataLoader(dataset, batch_size=256, shuffle=False, num_workers=0)
    featureExtractor.eval()

    allEmbeddings = []

    with torch.no_grad():
        for images, _ in loader:
            images = images.to(device)
            features = featureExtractor(images)
            allEmbeddings.append(features.cpu())

    return torch.cat(allEmbeddings, dim=0)
```

In [ ]:

```
def runFramework1TpcrpLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=3):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    warmStartIndices = random.sample(unlabeledIndices, initialBudget)
    warmStartModel = trainSupervisedModel(trainSet, warmStartIndices, numEpochs=5)

    warmFeatureExtractor = ResNet18FeatureExtractor(warmStartModel).to(device)
    warmTrainEmbeddings = extractResNetEmbeddings(warmFeatureExtractor, trainSet)

    initialSelected = selectInitialTpcrp(warmTrainEmbeddings, unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSupervisedModel(trainSet, labeledIndices, numEpochs=5)
        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        featureExtractor = ResNet18FeatureExtractor(model).to(device)
        trainEmbeddings = extractResNetEmbeddings(featureExtractor, trainSet)

        newSelected = selectTpcrpBatch(
            trainEmbeddings,
            labeledIndices,
            unlabeledIndices,
```

```
            batchSize
        )

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [ ]:
```
def runFramework1RandomLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=3):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSupervisedModel(trainSet, labeledIndices, numEpochs=5)
        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })


        if roundNum == numRounds:
            break

        newSelected = selectRandomBatch(unlabeledIndices, batchSize)
        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [59]:
```
def selectMarginFromProbabilities(probabilities, candidateIndices, batchSize):
    sortedProbs, _ = torch.sort(probabilities, dim=1, descending=True)
    margins = sortedProbs[:, 0] - sortedProbs[:, 1]
    chosenPositions = torch.argsort(margins)[:batchSize]
    selectedIndices = [candidateIndices[pos] for pos in chosenPositions.tolist()]
    return selectedIndices
```

In [ ]:
```
def selectEntropyFromProbabilities(probabilities, candidateIndices, batchSize):
    entropy = -(probabilities * torch.log(probabilities + 1e-12)).sum(dim=1)
    chosenPositions = torch.argsort(entropy, descending=True)[:batchSize]
    selectedIndices = [candidateIndices[pos] for pos in chosenPositions.tolist()]
    return selectedIndices
```

In [58]:
```
def runFramework2MarginLoop(trainEmbeddings, trainLabels, testEmbeddings, testLabels, initialBudget=10, batchSi
    labeledIndices = []
    unlabeledIndices = list(range(len(trainEmbeddings)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        linearClassifier = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)
        accuracy = evaluateLinearClassifier(linearClassifier, testEmbeddings, testLabels)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework2Probabilities(linearClassifier, trainEmbeddings, unlabeledIndices)
        newSelected = selectMarginFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [65]:
```
def runFramework2EntropyLoop(trainEmbeddings, trainLabels, testEmbeddings, testLabels, initialBudget=10, batchS
    labeledIndices = []
    unlabeledIndices = list(range(len(trainEmbeddings)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        linearClassifier = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)
        accuracy = evaluateLinearClassifier(linearClassifier, testEmbeddings, testLabels)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework2Probabilities(linearClassifier, trainEmbeddings, unlabeledIndices)
        newSelected = selectEntropyFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [60]:
```
def getFramework1Probabilities(model, trainSet, unlabeledIndices):
    unlabeledSubset = Subset(trainSet, unlabeledIndices)
    unlabeledLoader = DataLoader(unlabeledSubset, batch_size=128, shuffle=False, num_workers=0)

    model.eval()
    allProbabilities = []

    with torch.no_grad():
        for images, _ in unlabeledLoader:
            images = images.to(device)
            logits = model(images)
            probabilities = torch.softmax(logits, dim=1)
            allProbabilities.append(probabilities.cpu())

    return torch.cat(allProbabilities, dim=0)
```

In [61]:
```
def runFramework1MarginLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSupervisedModel(trainSet, labeledIndices, numEpochs=2)
        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework1Probabilities(model, trainSet, unlabeledIndices)
        newSelected = selectMarginFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [62]:
```
def runFramework1EntropyLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []
```

```
    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSupervisedModel(trainSet, labeledIndices, numEpochs=2)
        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework1Probabilities(model, trainSet, unlabeledIndices)
        newSelected = selectEntropyFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [87]:

```
def trainSemiSupervisedModel(trainSet, labeledIndices, unlabeledIndices, numEpochs=1, confidenceThreshold=0.9):
    labeledSubset = Subset(trainSet, labeledIndices)
    unlabeledSubset = Subset(trainSet, unlabeledIndices)

    labeledBatchSize = min(8, len(labeledSubset)) if len(labeledSubset) > 0 else 1
    unlabeledBatchSize = 32

    labeledLoader = DataLoader(
        labeledSubset,
        batch_size=labeledBatchSize,
        shuffle=True,
        num_workers=0,
        drop_last=False
    )

    unlabeledLoader = DataLoader(
        unlabeledSubset,
        batch_size=unlabeledBatchSize,
        shuffle=True,
        num_workers=0,
        drop_last=True
    )

    model = ResNet18Classifier(numClasses=10).to(device)
    lossFn = nn.CrossEntropyLoss()
    optimizer = optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-4)

    for epoch in range(numEpochs):
        model.train()

        for images, labels in labeledLoader:
            if images.size(0) < 2:
                continue

            images = images.to(device)
            labels = labels.to(device)

            optimizer.zero_grad()
            logits = model(images)
            supervisedLoss = lossFn(logits, labels)
            supervisedLoss.backward()
            optimizer.step()


        for images, _ in unlabeledLoader:
            if images.size(0) < 2:
                continue

            images = images.to(device)

            with torch.no_grad():
                model.eval()
                logits = model(images)
                probabilities = torch.softmax(logits, dim=1)
                maxProbs, pseudoLabels = torch.max(probabilities, dim=1)
```

```
            mask = maxProbs >= confidenceThreshold
            if mask.sum().item() < 2:
                model.train()
                continue

            confidentImages = images[mask]
            confidentPseudoLabels = pseudoLabels[mask]

            if confidentImages.size(0) < 2:
                model.train()
                continue

            model.train()
            optimizer.zero_grad()
            confidentLogits = model(confidentImages)
            unsupervisedLoss = lossFn(confidentLogits, confidentPseudoLabels)
            unsupervisedLoss.backward()
            optimizer.step()

        print(f"Epoch {epoch + 1}/{numEpochs} done")

    return model
```

In [88]:

```
def runFramework3RandomLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSemiSupervisedModel(
            trainSet,
            labeledIndices,
            unlabeledIndices,
            numEpochs=1,
            confidenceThreshold=0.9
        )

        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        newSelected = selectRandomBatch(unlabeledIndices, batchSize)
        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [85]:

```
def runFramework3TpcrpLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    warmStartIndices = random.sample(unlabeledIndices, initialBudget)
    warmModel = trainSemiSupervisedModel(
        trainSet,
        warmStartIndices,
        [idx for idx in unlabeledIndices if idx not in set(warmStartIndices)],
        numEpochs=1,
        confidenceThreshold=0.9
    )

    warmFeatureExtractor = ResNet18FeatureExtractor(warmModel).to(device)
    warmTrainEmbeddings = extractResNetEmbeddings(warmFeatureExtractor, trainSet)

    initialSelected = selectInitialTpcrp(warmTrainEmbeddings, unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSemiSupervisedModel(
            trainSet,
```

```
            labeledIndices,
            unlabeledIndices,
            numEpochs=1,
            confidenceThreshold=0.9
        )

        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        featureExtractor = ResNet18FeatureExtractor(model).to(device)
        trainEmbeddings = extractResNetEmbeddings(featureExtractor, trainSet)

        newSelected = selectTpcrpBatch(
            trainEmbeddings,
            labeledIndices,
            unlabeledIndices,
            batchSize
        )

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [75]:

```
def getFramework3Probabilities(model, trainSet, unlabeledIndices):
    unlabeledSubset = Subset(trainSet, unlabeledIndices)
    unlabeledLoader = DataLoader(unlabeledSubset, batch_size=128, shuffle=False, num_workers=0)

    model.eval()
    allProbabilities = []

    with torch.no_grad():
        for images, _ in unlabeledLoader:
            images = images.to(device)
            logits = model(images)
            probabilities = torch.softmax(logits, dim=1)
            allProbabilities.append(probabilities.cpu())

    return torch.cat(allProbabilities, dim=0)
```

In [90]:

```
def runFramework3MarginLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSemiSupervisedModel(
            trainSet,
            labeledIndices,
            unlabeledIndices,
            numEpochs=1,
            confidenceThreshold=0.9
        )

        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework3Probabilities(model, trainSet, unlabeledIndices)
        newSelected = selectMarginFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
```

```
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [91]:
```
def runFramework3EntropyLoop(trainSet, testLoader, initialBudget=10, batchSize=10, numRounds=5):
    labeledIndices = []
    unlabeledIndices = list(range(len(trainSet)))
    results = []

    initialSelected = selectRandomBatch(unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        model = trainSemiSupervisedModel(
            trainSet,
            labeledIndices,
            unlabeledIndices,
            numEpochs=1,
            confidenceThreshold=0.9
        )

        accuracy = evaluateClassifier(model, testLoader)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework3Probabilities(model, trainSet, unlabeledIndices)
        newSelected = selectEntropyFromProbabilities(probabilities, unlabeledIndices, batchSize)

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [70]:
```
framework1TpcrpResults = runFramework1TpcrpLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework1RandomResults = runFramework1RandomLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework1MarginResults = runFramework1MarginLoop(
    trainSet, testLoader,
    initialBudget=10, batchSize=10, numRounds=5
)

framework1EntropyResults = runFramework1EntropyLoop(
    trainSet, testLoader,
    initialBudget=10, batchSize=10, numRounds=5
)
print("Framework 1 TPCRP Results:")
for row in framework1TpcrpResults:
    print(row)

print("\nFramework 1 Random Results:")
for row in framework1RandomResults:
    print(row)

print("\nFramework 1 Margin Results:")
for row in framework1MarginResults:
    print(row)

print("\nFramework 1 Entropy Results:")
for row in framework1EntropyResults:
    print(row)
```

```
Epoch 1/5, Loss: 2.7049, Train Acc: 0.1000
Epoch 2/5, Loss: 0.6164, Train Acc: 1.0000
Epoch 3/5, Loss: 0.0318, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0088, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0050, Train Acc: 1.0000
Epoch 1/5, Loss: 2.4183, Train Acc: 0.1000
Epoch 2/5, Loss: 0.7534, Train Acc: 0.9000
Epoch 3/5, Loss: 0.0471, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0114, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0053, Train Acc: 1.0000
Epoch 1/5, Loss: 2.7982, Train Acc: 0.0500
Epoch 2/5, Loss: 0.6905, Train Acc: 0.9500
Epoch 3/5, Loss: 0.0430, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0141, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0089, Train Acc: 1.0000
Epoch 1/5, Loss: 2.8447, Train Acc: 0.0000
Epoch 2/5, Loss: 0.6694, Train Acc: 0.9667
Epoch 3/5, Loss: 0.0787, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0220, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0106, Train Acc: 1.0000
Epoch 1/5, Loss: 2.3738, Train Acc: 0.1500
Epoch 2/5, Loss: 0.7671, Train Acc: 0.8750
Epoch 3/5, Loss: 0.2006, Train Acc: 0.9250
Epoch 4/5, Loss: 0.1572, Train Acc: 0.9750
Epoch 5/5, Loss: 0.1358, Train Acc: 0.9750
Epoch 1/5, Loss: 2.7051, Train Acc: 0.0600
Epoch 2/5, Loss: 0.8267, Train Acc: 0.9200
Epoch 3/5, Loss: 0.1322, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0287, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0171, Train Acc: 1.0000
Epoch 1/5, Loss: 2.6332, Train Acc: 0.0667
Epoch 2/5, Loss: 0.8995, Train Acc: 0.8500
Epoch 3/5, Loss: 0.1895, Train Acc: 0.9833
Epoch 4/5, Loss: 0.0419, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0205, Train Acc: 1.0000
Epoch 1/5, Loss: 2.6049, Train Acc: 0.1000
Epoch 2/5, Loss: 0.4506, Train Acc: 1.0000
Epoch 3/5, Loss: 0.0242, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0068, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0032, Train Acc: 1.0000
Epoch 1/5, Loss: 2.6419, Train Acc: 0.0500
Epoch 2/5, Loss: 0.3738, Train Acc: 1.0000
Epoch 3/5, Loss: 0.0264, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0088, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0048, Train Acc: 1.0000
Epoch 1/5, Loss: 2.3564, Train Acc: 0.1333
Epoch 2/5, Loss: 0.4996, Train Acc: 0.9333
Epoch 3/5, Loss: 0.0452, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0129, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0070, Train Acc: 1.0000
Epoch 1/5, Loss: 2.4515, Train Acc: 0.0500
Epoch 2/5, Loss: 0.6090, Train Acc: 1.0000
Epoch 3/5, Loss: 0.0611, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0210, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0124, Train Acc: 1.0000
Epoch 1/5, Loss: 2.6483, Train Acc: 0.0600
Epoch 2/5, Loss: 0.6258, Train Acc: 0.9600
Epoch 3/5, Loss: 0.0803, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0265, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0124, Train Acc: 1.0000
Epoch 1/5, Loss: 2.6847, Train Acc: 0.0833
Epoch 2/5, Loss: 0.7121, Train Acc: 0.9000
Epoch 3/5, Loss: 0.1153, Train Acc: 1.0000
Epoch 4/5, Loss: 0.0310, Train Acc: 1.0000
Epoch 5/5, Loss: 0.0144, Train Acc: 1.0000
Epoch 1/2, Loss: 2.3758, Train Acc: 0.2000
Epoch 2/2, Loss: 0.5240, Train Acc: 0.9000
Epoch 1/2, Loss: 2.2457, Train Acc: 0.1500
Epoch 2/2, Loss: 0.4606, Train Acc: 1.0000
Epoch 1/2, Loss: 2.6098, Train Acc: 0.1000
Epoch 2/2, Loss: 0.5281, Train Acc: 0.9667
Epoch 1/2, Loss: 2.7396, Train Acc: 0.1250
Epoch 2/2, Loss: 0.7340, Train Acc: 0.9000
Epoch 1/2, Loss: 2.4492, Train Acc: 0.1000
Epoch 2/2, Loss: 0.5166, Train Acc: 0.9800
Epoch 1/2, Loss: 2.5584, Train Acc: 0.0667
Epoch 2/2, Loss: 0.6274, Train Acc: 0.9667
Epoch 1/2, Loss: 2.3613, Train Acc: 0.2000
Epoch 2/2, Loss: 0.3875, Train Acc: 1.0000
Epoch 1/2, Loss: 2.7470, Train Acc: 0.0500
Epoch 2/2, Loss: 0.5260, Train Acc: 1.0000
Epoch 1/2, Loss: 2.8406, Train Acc: 0.0667
Epoch 2/2, Loss: 0.5117, Train Acc: 0.9667
```

```
Epoch 1/2, Loss: 2.4305, Train Acc: 0.1750
Epoch 2/2, Loss: 0.5463, Train Acc: 1.0000
Epoch 1/2, Loss: 2.5722, Train Acc: 0.0600
Epoch 2/2, Loss: 0.6174, Train Acc: 0.9600
Epoch 1/2, Loss: 2.6223, Train Acc: 0.0167
Epoch 2/2, Loss: 0.7397, Train Acc: 0.9833
Framework 1 TPCRP Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.1053}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1146}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1323}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1064}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.1003}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1503}

Framework 1 Random Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.0971}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1001}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1044}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1513}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.1011}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1}

Framework 1 Margin Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.1189}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1254}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1032}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1196}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.122}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1031}

Framework 1 Entropy Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.0984}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.0987}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1189}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1226}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.1402}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1146}
```

In [113…

```python
framework2TpcrpResults = runFramework2TpcrpLoop(
    trainEmbeddings,
    trainLabels,
    testEmbeddings,
    testLabels,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework2RandomResults = runFramework2RandomLoop(
    trainEmbeddings,
    trainLabels,
    testEmbeddings,
    testLabels,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework2MarginResults = runFramework2MarginLoop(
    trainEmbeddings, trainLabels, testEmbeddings, testLabels,
    initialBudget=10, batchSize=10, numRounds=5
)

framework2EntropyResults = runFramework2EntropyLoop(
    trainEmbeddings, trainLabels, testEmbeddings, testLabels,
    initialBudget=10, batchSize=10, numRounds=5
)

print("Framework 2 TPCRP Results:")
for row in framework2TpcrpResults:
    print(row)

print("\n Framework 2 Random Results:")
for row in framework2RandomResults:
    print(row)

print("Framework 2 Margin Results:")
for row in framework2MarginResults:
    print(row)

print("Framework 2 Entropy Results:")
for row in framework2EntropyResults:
    print(row)
```

```
Framework 2 TPCRP Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.536300003528595}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.6424000263214111}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.6567000150680542}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.6743999719619751}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.6682000160217285}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.6801999807357788}

 Framework 2 Random Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.39719998836517334}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.5859000086784363}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.628000020980835}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.6308000087738037}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.6582000255584717}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.6657000184059143}
Framework 2 Margin Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.3666999936103821}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.5803999900817871}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.6317999958992004}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.647599995136261}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.6658999919891357}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.6844000220298767}
Framework 2 Entropy Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.4260999858379364}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.4593000113964081}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.508400022983551}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.5497999787330627}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.5612999796867371}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.631600022315979}
```

In [93]:

```python
framework3TpcrpResults = runFramework3TpcrpLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework3RandomResults = runFramework3RandomLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)
framework3MarginResults = runFramework3MarginLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

framework3EntropyResults = runFramework3EntropyLoop(
    trainSet,
    testLoader,
    initialBudget=10,
    batchSize=10,
    numRounds=5
)

print("Framework 3 TPCRP Results:")
for row in framework3TpcrpResults:
    print(row)

print("\nFramework 3 Random Results:")
for row in framework3RandomResults:
    print(row)

print("Framework 3 Margin Results:")
for row in framework3MarginResults:
    print(row)

print("\nFramework 3 Entropy Results:")
for row in framework3EntropyResults:
    print(row)
```

```
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
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Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Epoch 1/1 done
Framework 3 TPCRP Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.1187}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1037}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1329}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1149}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.0933}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1001}

Framework 3 Random Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.0857}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1072}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.146}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.0863}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.0975}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.0998}
Framework 3 Margin Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.1005}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1045}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1021}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.1045}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.1}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1011}

Framework 3 Entropy Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.0994}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.1094}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.1}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.0789}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.1}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.1001}
```

In [94]:

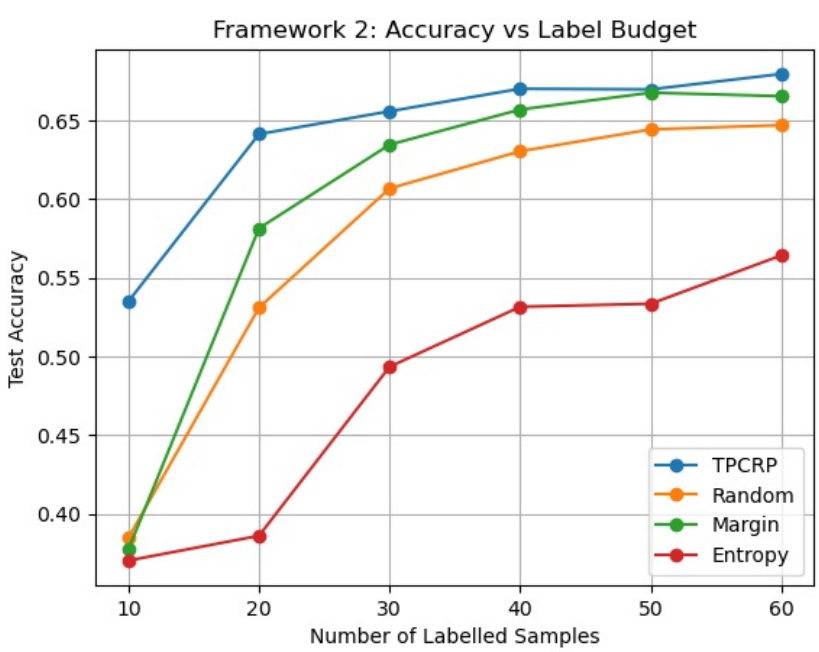
```
framework1TpcrpDf = pd.DataFrame(framework1TpcrpResults)
framework1TpcrpDf["method"] = "TPCRP"
framework1TpcrpDf["framework"] = "Framework1"

framework1RandomDf = pd.DataFrame(framework1RandomResults)
framework1RandomDf["method"] = "Random"
framework1RandomDf["framework"] = "Framework1"

framework1MarginDf = pd.DataFrame(framework1MarginResults)
framework1MarginDf["method"] = "Margin"
framework1MarginDf["framework"] = "Framework1"

framework1EntropyDf = pd.DataFrame(framework1EntropyResults)
framework1EntropyDf["method"] = "Entropy"
framework1EntropyDf["framework"] = "Framework1"

framework2TpcrpDf = pd.DataFrame(framework2TpcrpResults)
framework2TpcrpDf["method"] = "TPCRP"
framework2TpcrpDf["framework"] = "Framework2"

framework2RandomDf = pd.DataFrame(framework2RandomResults)
framework2RandomDf["method"] = "Random"
framework2RandomDf["framework"] = "Framework2"

framework2MarginDf = pd.DataFrame(framework2MarginResults)
framework2MarginDf["method"] = "Margin"
framework2MarginDf["framework"] = "Framework2"
```

```
framework2EntropyDf = pd.DataFrame(framework2EntropyResults)
framework2EntropyDf["method"] = "Entropy"
framework2EntropyDf["framework"] = "Framework2"

framework3TpcrpDf = pd.DataFrame(framework3TpcrpResults)
framework3TpcrpDf["method"] = "TPCRP"
framework3TpcrpDf["framework"] = "Framework3"

framework3RandomDf = pd.DataFrame(framework3RandomResults)
framework3RandomDf["method"] = "Random"
framework3RandomDf["framework"] = "Framework3"

framework3MarginDf = pd.DataFrame(framework3MarginResults)
framework3MarginDf["method"] = "Margin"
framework3MarginDf["framework"] = "Framework3"

framework3EntropyDf = pd.DataFrame(framework3EntropyResults)
framework3EntropyDf["method"] = "Entropy"
framework3EntropyDf["framework"] = "Framework3"

resultsDf = pd.concat([framework3TpcrpDf, framework3RandomDf, framework3MarginDf, framework3EntropyDf, framework
resultsDf.to_csv("frameworkResults.csv", index=False)
print(resultsDf)
```

```
    round  numLabeled  accuracy   method   framework
0       0          10    0.1187    TPCRP  Framework3
1       1          20    0.1037    TPCRP  Framework3
2       2          30    0.1329    TPCRP  Framework3
3       3          40    0.1149    TPCRP  Framework3
4       4          50    0.0933    TPCRP  Framework3
..    ...         ...       ...      ...         ...
67      1          20    0.0987  Entropy  Framework1
68      2          30    0.1189  Entropy  Framework1
69      3          40    0.1226  Entropy  Framework1
70      4          50    0.1402  Entropy  Framework1
71      5          60    0.1146  Entropy  Framework1

[72 rows x 5 columns]
```

In [101…

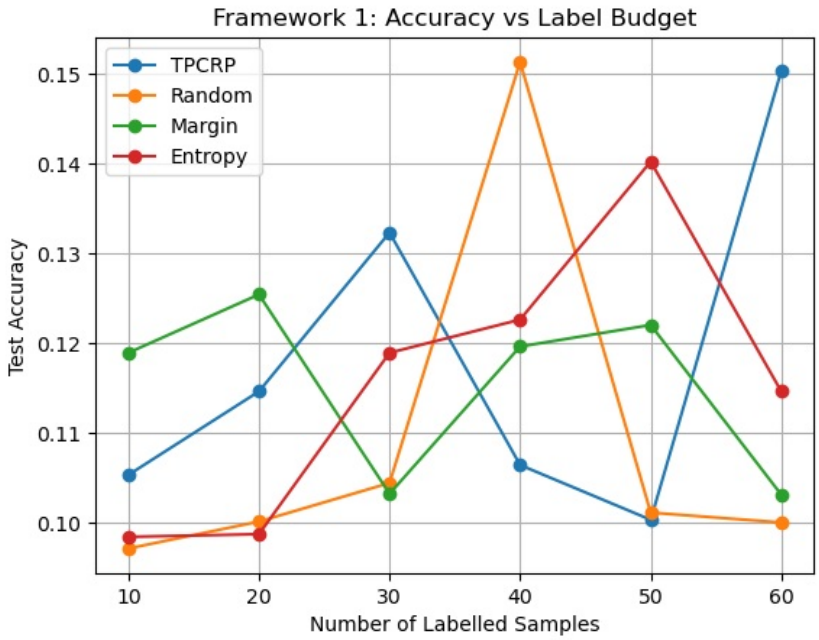
```
allDf = pd.read_csv("frameworkResults.csv")

framework2Df = allDf[allDf["framework"] == "Framework2"]

for method in framework2Df["method"].unique():
    methodDf = framework2Df[framework2Df["method"] == method]
    plt.plot(methodDf["numLabeled"], methodDf["accuracy"], marker="o", label=method)

plt.xlabel("Number of Labelled Samples")
plt.ylabel("Test Accuracy")
plt.title("Framework 2: Accuracy vs Label Budget")
plt.legend()
plt.grid(True)
plt.show()
```



In [99]:

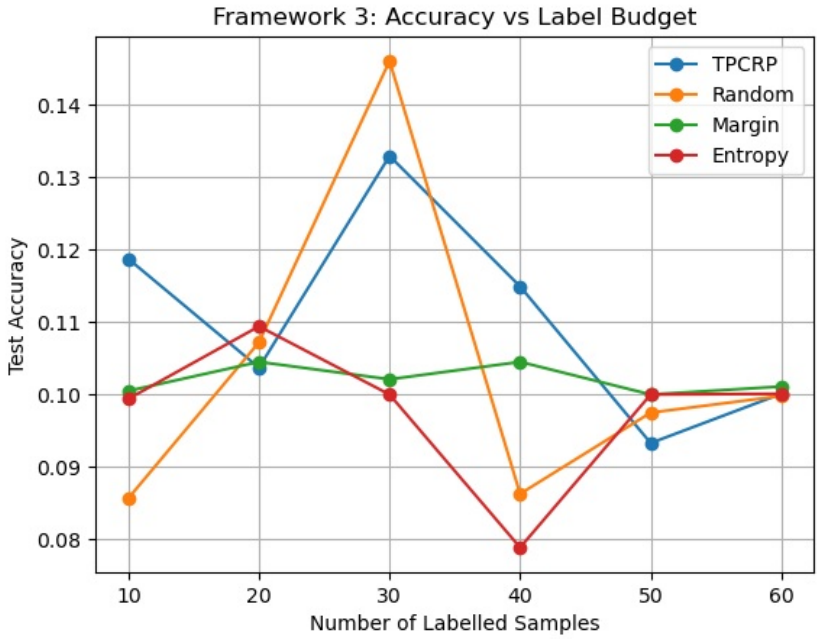
```
allDf = pd.read_csv("frameworkResults.csv")
```

```
framework1Df = allDf[allDf["framework"] == "Framework1"]

for method in framework1Df["method"].unique():
    methodDf = framework1Df[framework1Df["method"] == method]
    plt.plot(methodDf["numLabeled"], methodDf["accuracy"], marker="o", label=method)

plt.xlabel("Number of Labelled Samples")
plt.ylabel("Test Accuracy")
plt.title("Framework 1: Accuracy vs Label Budget")
plt.legend()
plt.grid(True)
plt.show()
```



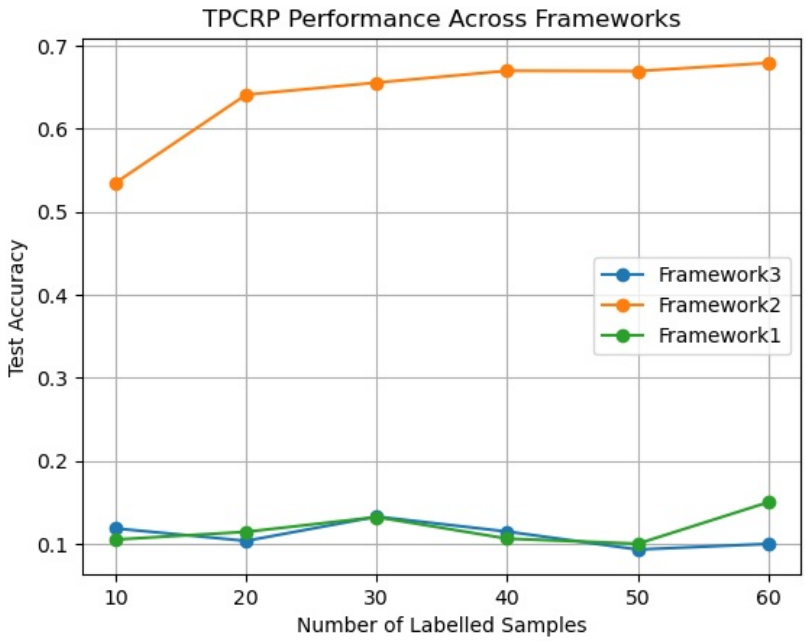
In [100…

```
allDf = pd.read_csv("frameworkResults.csv")

framework3Df = allDf[allDf["framework"] == "Framework3"]

for method in framework3Df["method"].unique():
    methodDf = framework3Df[framework3Df["method"] == method]
    plt.plot(methodDf["numLabeled"], methodDf["accuracy"], marker="o", label=method)

plt.xlabel("Number of Labelled Samples")
plt.ylabel("Test Accuracy")
plt.title("Framework 3: Accuracy vs Label Budget")
plt.legend()
plt.grid(True)
plt.show()
```



In [102…

```
tpcrpDf = allDf[allDf["method"] == "TPCRP"]
for framework in tpcrpDf["framework"].unique():
    frameworkDf = tpcrpDf[tpcrpDf["framework"] == framework]
    plt.plot(frameworkDf["numLabeled"], frameworkDf["accuracy"], marker="o", label=framework)

plt.xlabel("Number of Labelled Samples")
plt.ylabel("Test Accuracy")
plt.title("TPCRP Performance Across Frameworks")
plt.legend()
plt.grid(True)
plt.show()
```



In [103…

```
def minMaxNormalize(values):
    values = np.array(values, dtype=np.float32)
    if values.max() - values.min() < 1e-8:
        return np.ones_like(values)
    return (values - values.min()) / (values.max() - values.min())
```

In [104…

```
def computeEntropyScores(probabilities):
    entropy = -(probabilities * torch.log(probabilities + 1e-12)).sum(dim=1)
    return entropy.cpu().numpy()
```

In [109…

```
def selectTpcrpEntropyHybrid(trainEmbeddings, labeledIndices, unlabeledIndices, batchSize, probabilities, alpha
    labeledIndices = [int(idx) for idx in labeledIndices]
    unlabeledIndices = [int(idx) for idx in unlabeledIndices]

    allIndices = labeledIndices + unlabeledIndices
    allEmbeddings = trainEmbeddings[allIndices].cpu().numpy()

    numClusters = len(labeledIndices) + batchSize
    kmeans = KMeans(n_clusters=numClusters, random_state=42, n_init=10)
    clusterLabels = kmeans.fit_predict(allEmbeddings)

    clusterToMembers = {}
    for pos, clusterId in enumerate(clusterLabels):
        globalIdx = allIndices[pos]
        clusterToMembers.setdefault(clusterId, []).append(globalIdx)

    uncoveredClusters = []
    for clusterId, members in clusterToMembers.items():
        hasLabeled = any(idx in labeledIndices for idx in members)
        if not hasLabeled:
            uncoveredClusters.append((clusterId, members))

    uncoveredClusters.sort(key=lambda x: len(x[1]), reverse=True)
    chosenClusters = uncoveredClusters[:batchSize]

    entropyScoresAll = computeEntropyScores(probabilities)
    probIndexMap = {idx: pos for pos, idx in enumerate(unlabeledIndices)}

    selectedIndices = []

    for clusterId, members in chosenClusters:
        unlabeledMembers = [idx for idx in members if idx in unlabeledIndices]
```

```
        if len(unlabeledMembers) == 0:
            continue

        clusterEmbeddings = trainEmbeddings[unlabeledMembers]
        typicalityScores = computeTypicality(clusterEmbeddings)

        memberEntropy = np.array([
            entropyScoresAll[probIndexMap[idx]] for idx in unlabeledMembers
        ])

        typicalityNorm = minMaxNormalize(typicalityScores)
        entropyNorm = minMaxNormalize(memberEntropy)

        hybridScores = (typicalityNorm ** alpha) * (entropyNorm ** (1 - alpha))

        bestLocal = int(np.argmax(hybridScores))
        bestGlobal = unlabeledMembers[bestLocal]

        selectedIndices.append(bestGlobal)

    return selectedIndices
```

In [106…

```
def runFramework2HybridLoop(trainEmbeddings, trainLabels, testEmbeddings, testLabels, initialBudget=10, batchSi
    labeledIndices = []
    unlabeledIndices = list(range(len(trainEmbeddings)))
    results = []

    initialSelected = selectInitialTpcrp(trainEmbeddings, unlabeledIndices, initialBudget)
    labeledIndices.extend(initialSelected)
    unlabeledIndices = [idx for idx in unlabeledIndices if idx not in set(initialSelected)]

    for roundNum in range(numRounds + 1):
        linearClassifier = trainLinearClassifier(trainEmbeddings, trainLabels, labeledIndices)
        accuracy = evaluateLinearClassifier(linearClassifier, testEmbeddings, testLabels)

        results.append({
            "round": roundNum,
            "numLabeled": len(labeledIndices),
            "accuracy": accuracy
        })

        if roundNum == numRounds:
            break

        probabilities = getFramework2Probabilities(linearClassifier, trainEmbeddings, unlabeledIndices)

        newSelected = selectTpcrpEntropyHybrid(
            trainEmbeddings,
            labeledIndices,
            unlabeledIndices,
            batchSize,
            probabilities,
            alpha=alpha
        )

        labeledIndices.extend(newSelected)
        newSelectedSet = set(newSelected)
        unlabeledIndices = [idx for idx in unlabeledIndices if idx not in newSelectedSet]

    return results
```

In [111…

```
framework2HybridResults = runFramework2HybridLoop(
    trainEmbeddings,
    trainLabels,
    testEmbeddings,
    testLabels,
    initialBudget=10,
    batchSize=10,
    numRounds=5,
    alpha=0.7
)

print("Framework 2 TPCRP-Entropy Results:")
for row in framework2HybridResults:
    print(row)
```

```
Framework 2 TPCRP-Entropy Results:
{'round': 0, 'numLabeled': 10, 'accuracy': 0.5336999893188477}
{'round': 1, 'numLabeled': 20, 'accuracy': 0.6172000169754028}
{'round': 2, 'numLabeled': 30, 'accuracy': 0.6600000262260437}
{'round': 3, 'numLabeled': 40, 'accuracy': 0.6679999828338623}
{'round': 4, 'numLabeled': 50, 'accuracy': 0.683899998664856}
{'round': 5, 'numLabeled': 60, 'accuracy': 0.6934999823570251}
```

In [112…

```
framework2HybridDf = pd.DataFrame(framework2HybridResults)
framework2HybridDf["method"] = "TPCRP-Entropy"
framework2HybridDf["framework"] = "Framework2"

framework2TpcrpDf = framework2Df[framework2Df["method"] == "TPCRP"].copy()
framework2RandomDf = framework2Df[framework2Df["method"] == "Random"].copy()
framework2MarginDf = framework2Df[framework2Df["method"] == "Margin"].copy()
framework2EntropyDf = framework2Df[framework2Df["method"] == "Entropy"].copy()
updatedFramework2Df = pd.concat([framework2TpcrpDf, framework2RandomDf, framework2MarginDf, framework2EntropyDf
updatedFramework2Df.to_csv("framework2UpdatedResults.csv", index=False)
```

8.2 GitHub Link

https://github.com/jodieorogun/5CCSAMLF-coursework2